കണ്ണൂർ ജില്ലാ പഞ്ചായത്ത്

പൊതുവിദ്യാഭ്യാസ
വകുപ്പ്- കേരളം

ഡയറ്റ് കണ്ണൂർ

समग्र शिक्षा के रल
Samagra Shiksha Kerala
സമഗ്ര ശിക്ഷാ കേരളം



എസ് എസ് എൽ സി
വിദ്യാർത്ഥികൾക്കുള്ള
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ബയോളജി

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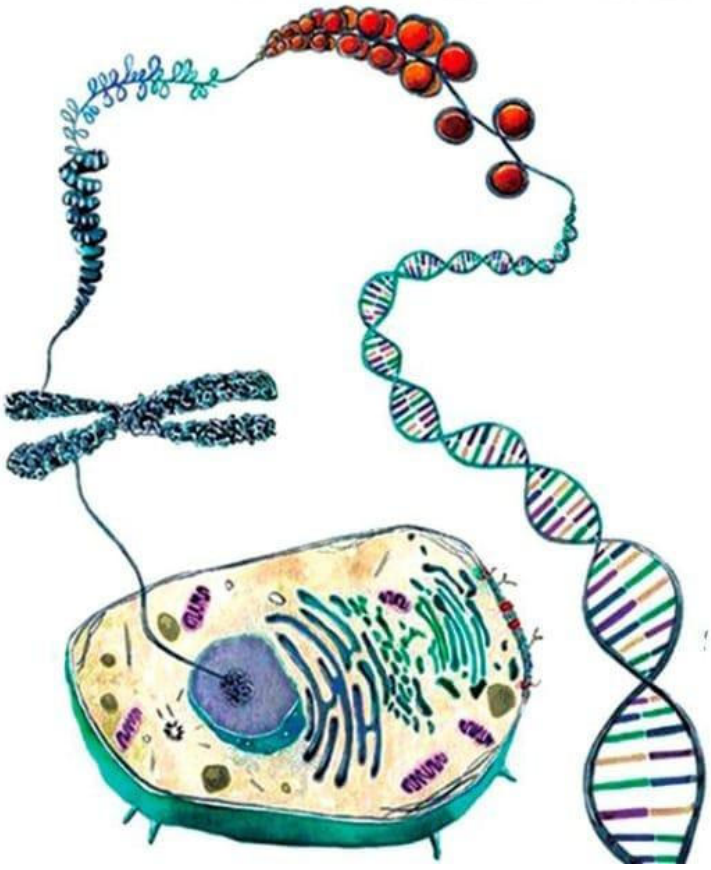
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SMILE 2025-26

BIOLOGY



ENGLISH MEDIUM

PREFACE

During the last academic year, students from the Kannur district achieved excellent success at the state level in the 10th standard examinations. It is the duty of each one of us to maintain and further improve this success in the coming years as well. Recognizing its importance, the Kannur District Panchayat has decided to implement the 'Smile' project this year in a timely manner. We have attempted to prepare a material incorporating activities that will enable students to achieve high grades in Biology, which holds significant importance among science subjects. To achieve the learning objectives in Biology, students need to go through learning experiences and activities. For this, concepts from the textbook have been explained and model questions have been prepared. The aim of this initiative is to guide students to a high standard, while working within the time allotted for science subjects.

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Contents

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2 .Paths of Evolution ……...15-36

3 .Behind Sensations ……...37-54

CHAPTER : 1

GENETICS OF LIFE

Scientists associated with gene editing

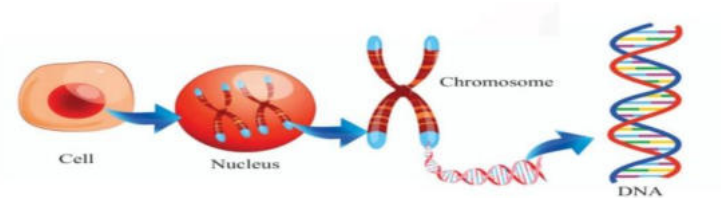
- Winners of the 2020 Nobel Prize in Chemistry for their contributions in the field of gene editing.

➢ Emmanuelle Charpentier.

➢ Jennifer A Doudna.

- CRISPR-Cas-9: A technology that aids in the gene editing process.

The Location of DNA- DNA is found in the chromosomes inside the nucleus of the cell.



Scientists associated with the structure of DNA

➢ Rosalind Franklin - X-ray diffraction (Photo 51)

➢ Maurice Wilkins – Study of X-Ray

➢ James Watson and Francis Crick-double helix model of DNA

Evaluation

Statement 1 - The CRISPR-Cas9 technology was discovered by Emmanuelle Charpentier and Jennifer A. Doudna together.

Statement 2 - James Watson and Francis Crick discovered the double helix model of DNA.

a) Statement 1 correct, 2 incorrect b) Statement 1 incorrect, 2 correct

c) Statement 1 and 2 correct d) Both statements are incorrect

Ans:C

II. Complete the flow chart

Cell → …. A……. → Chromosome → B…..

Ans. A. Nucleus. B. DNA

Structure of DNA

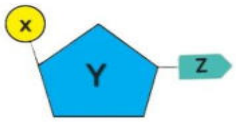
- ➢ DNA has a double helix structure.
- ➢ The strands of DNA are formed by deoxyribose sugar and phosphate .
- ➢ Rungs are formed by the pairing of nitrogen bases.
- ➢ DNA contains the nitrogenous bases adenine, thymine, guanine, and cytosine.
- ➢ Adenine pairs with thymine and guanine with cytosine.

Nucleotide

- The basic building block of DNA.
- Each nucleotide is composed of a deoxyribose sugar, a phosphate group, and a nitrogen base.

Evaluation

I. Observe the illustration and write the answers to the questions.



a) The picure of the basic structural unit of DNA is given. Identify and write the name.

b) Write the function of the molecule X.

c) Write the names of Y and Z.

II. A table related to DNA is provided. Arrange it in the correct order and choose the right answer from the given options.

A	B

1, Number of strands in DNA	a) Nitrogen bases pair up to form
2, Different types of nitrogen bases	b) 2
3, Molecules in the nucleotide.	c) Adenine- Thymine, Guanine - Cytosine
4, steps are formed	d) Phosphate, nitrogen base deoxyribose

i)1.b, 2.c, 3.d, 4.a ii)1.b, 2.a, 3.c, 4.d

iii)1.d, 2.b, 3.c, 4.a

Ans:i

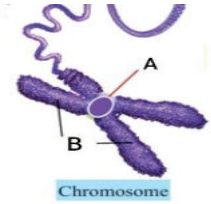
Size of DNA - The DNA in each chromosome is about 2 inches (5cm) long.

Structure of chromosome

- DNA and histone proteins are the primary components of a chromosome.
- Eight histone proteins join together to form a histone octamer.
- The DNA strands wind around this octamer to form a structure called nucleosome.
- The chromosomes are formed by packing and coiling numerous nucleosomes and recoiling the chains of nucleosomes.
- chromatids are the parts of a chromosome which are connected by centromere.

Evaluation

Observe figure and write answers to the questions.



1, Identify the parts A and B.

Ans. A- centromere B- chromatid

2, How the structure of the chromosome is formed by the combination of DNA and Proteins.

Human Chromosomes

- 44 somatic chromosomes + 2 sex chromosomes
- somatic chromosomes are chromosomes that control physical characteristics.
- There are 22 pairs of autosomes.
- Chromosomes responsible for sex determination.
- There are two types of sex-determining chromosomes **(X and Y).**

Evaluation

1, The Y chromosome of the father is important in sex determination. Do you agree with this statement? Explain with reasons?

2, complete the table

	Genetic constitution	Total Number of chromosomes	Number of somatic chromosomes	Number and type of sex chromosomes
Female	44+XX	P	44	Q
Male	R	46	S	1 pair: X and Y

Ans. P. 46, Q .1 pair XX, R.44+XY, S .44

Gene

- Gene is a specific sequence of nucleotides in DNA.
- Proteins, which are synthesized according to the instructions of genes, are responsible for the formation of characteristic features and for controlling metabolic activities.

RNA

- Like DNA, RNA is also made up of nucleotides.
- RNA is made up of a ribose sugar, a phosphate group, and a nitrogen base.

- The nitrogen bases in RNA are adenine, thymine, uracil, and cytosine.

Evaluation

Complete the table

	Number of strands	Type of sugar molecule	Nitrogen bases
DNA	1	……i….	Adenine - thymine, guanine - cytosine
RNA	2	Ribose	……...ii…….

Ans. i,Deoxyribose ii, Adenine-Uracil ,Thymine,-Cytosine

Protein Synthesis

Step 1: **Transcription**

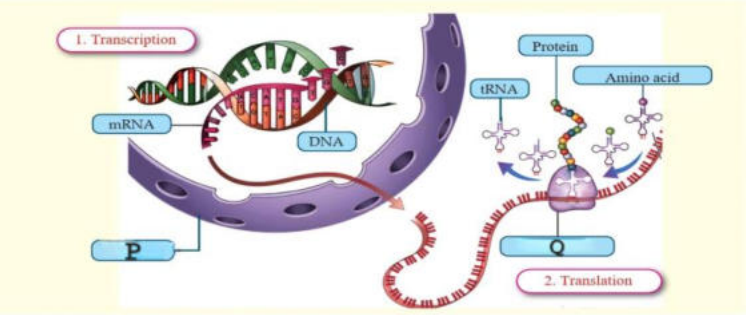
mRNA is formed from a specific nucleotide sequence (gene) in DNA with the help of enzymes. The mRNA contains messages for protein synthesis.

Step 2: **Translation**

- tRNAs (transfer RNA) carry specific amino acids to the ribosome based on message in the mRNA that has reached the ribosome from the nucleus.
- The rRNAs (ribosomal RNA), which are part of ribosomes combine amino acids to make protein

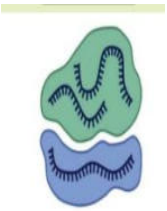
Evaluation

1. A diagram related to protein synthesis is given. Observe the diagram carefully and write answers to the questions below.

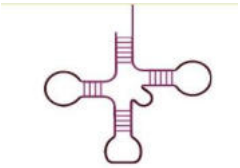


a, Identify and write the names of P and Q in the diagram.
Ans.P-Nucleus Q- Ribosome
b, Write the function of part Q
Ans.Protein Synthesis
c, Explain the process that takes place in the nucleus.

2. Observe the illustration and answer the questions given below.



X



Y

a, Identify and write the names of the different types of RNA labeled X and Y.
Ans.X,r RNA. Y,t RNA
b, Write one function of each X, Y.
X.Helps in the formation of bonds between amino acids.
Y.Carry specific amino acids to the ribosome based on message in the mRNA

Similarities and Differences in Characters

- **Heredity:** Heredity refers to the transmission of characteristics from parents to their offspring.
- **Variations** are characters expressed in offspring, that differ from their parents.
- **Genes** inherited from parents are responsible for both heredity and variations.

Gregor Johann Mendel - The Father of Genetics

Gregor Mendel's conclusions are known as the **Laws of Inheritance**.

- The laws of inheritance are the fundamental genetic framework to understand heredity and variation.
- **Evaluation**

Q) Genetics

Heredity

A

a, Identify and write 'A' from the given flowchart

b, Who is known as the Fther of Genetics?

Mendel's Experiments

- Gregor Mendel hypothesised that characters from parents are passed on to offsprings through certain factors that transmitted through gametes.
- It was only after Mendel's periods that ,these factors were discovered to be genes.
- A gene that determines a character can have different forms. These different forms of gene are **alleles.**
- The observable characteristics of an organism is phenotype.
- The genetic constitution responsible for these characteristics are genotype.

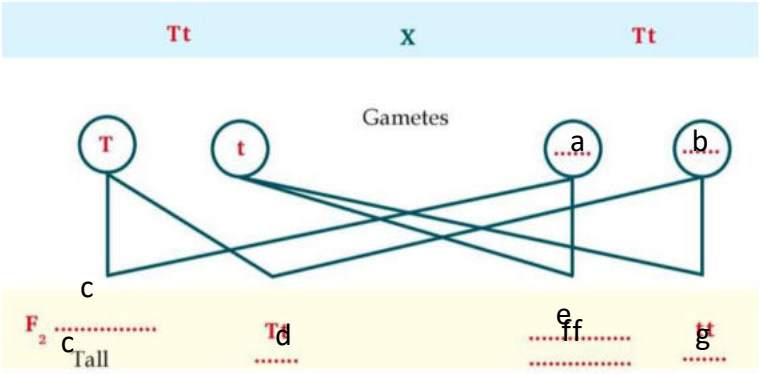
Evaluation

parent plants

TT X tt

...i... ...ii...

F_1 TR



Identify and write a,b,c,d,e,f,g

Dihybrid cross

- Gregor Mendel observed the inheritance of two contrasting traits of the same plant; this is known as a dihybrid cross.
- The traits considered were the plant's height and the shape of its seeds.
- In the dihybrid cross, the contrasting traits considered were tall and short for plant height, and round and wrinkled for seed shape.
- The parent plants in the dihybrid cross are the tall, round-seeded plant and the dwarf and wrinkled seeded plant.
- The genotypes of the parent plants are TTRR and ttrr. The first generation (F1) shows the traits of tall plant and round seeds(TtRr).
- Phenotypes observed in the second generation that differed in contrast to the parental plants (Tall plant with wrinkled seed,dwarf plant with round seed)
- The phenotypic ratio in the second generation (F2) is **9:3:3:1**.

The hypothesis that Mendel arrived at through the dihybrid Cross

- When two or more different traits are combined, each trait is inherited independently to the next generation without mixing each other.

<u>Evaluation</u>

1. TTRR X ttrr



TtRr

a, Complete i and ii in the given diagram.

i)TR ii)tr

b, What are the alleles in the gametes produced by the first generation?

TR,Tr,tR,tr

2.In Gregor Mentel's dihybrid cross, the first generation produced a tall, round-seeded plant.

Complete the illustration of its self-pollination.

	TR	Tr	tR	tr
TR	TTRR	……...i…….	TtRR	……..ii…….
Tr	…...iii…...	TTrr	……iv……..	Ttrr
tR	TtRR	…….v……..	ttRR	ttRr
tr	TtRr	Ttrr	TtRr	……..vi…...

a, Complete columns i,ii,iii,iv,v,vi.

Ans.I)TTRr, ii),TtRr, iii)TTRr iv)TtRr v)TtRr vi) ttrr

b, Phenotypic ratio in the second generation.

Ans.9:3:3:1

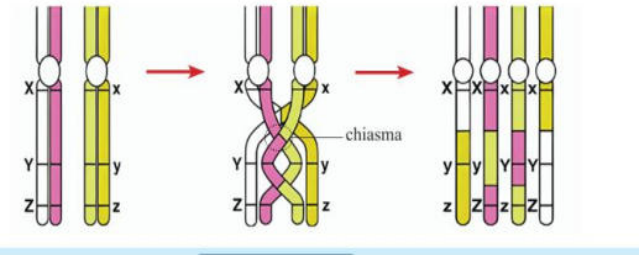
Non Mendelian Inheritance

A	B	C
Incomplete Dominance	A dominant allele cannot fully hide the allele of the recessive trait.	If a red flowered four o'clock plant is hybridized with a white flowered plant, the resulting offspring will have pink flowers.
Co-dominance	Both alleles exhibit their traits at the same time	Roan coat pattern, found on some cattle and horses
Multiple allelism	The gene that determines blood group in human beings has more than two alleles.	ABO blood group in humans
Polygenic inheritance	More than one gene controls the colour of the skin.	Difference in skin colour

Process that cause variations.

1. Crossing over

- Crossing over is the process that occurs during the first stage of meiosis.



a, Which process does the given diagram represent?

b, What is the role of this process in creating variations?

Ans.crossing over causes a recombination of alleles. This leads to the appearance of new traits in the offspring.

2.Mutation

- Mutation is the sudden heritable change in the genetic constitution of an organism.
- Errors that occur during DNA replication, certain chemicals, and radiation cause mutations.



Evaluation

a, Identify the given illustration and write the name of the process.
b, How does this process help in the formation of new species.
Ans.a)Mutation causes changes in genes.

b)These genes are transferred through generations which leads to variations in characters. Mutations play a crucial role in the process of evolution.

CHAPTER 2

PATHS OF EVOLUTION

Antimicrobial resistance

Bacteria that are resistant to antibiotics multiply and cause, diseases.This resistance caused due to mutations can be transmitted to their next generation. (multi-drug-resistant bacteria-superbugs)

Theory of evolution

1.Lamarckism or theory of inheritance of acquired characters(Lamarck)

The physical changes (inherited traits) that an organism acquires during its lifetime are passed on to the next generation.

Example:

When faced with food shortages, short-necked giraffes stretched their necks to eat leaves from higher branches, resulting in their necks becoming longer.

Evaluation:

1.Why is the theory of Lamarckism not considered scientifically accurate?

Scientists proved that acquired characters (variation) do not impart change in the genetic structure of organisms and therefore, are not inheritable.

2.Darwinism – the theory of natural selection (Charles Darwin)

- ➢ The ship Darwin sailed on - HMS Beagle
- ➢ Scientists who influenced Darwin - Thomas Malthus, Alfred Russel Wallace
- ➢ Darwin's book: ‘On the Origin of Species’.

Conduct a comparative study of Lamarckism and Darwinism.

Lamarckism	Darwinism
➢ Giraffes of earlier periods	➢ Giraffes with necks of

possessed short necks. ➢ Giraffes began to stretch their neck to obtain food. ➢ By continuous use giraffes with longer necks emerged ➢ Environment causes variations in organisms	varying length. ➢ Only the giraffes with longer necks survived in the competition for food ➢ Only the giraffes with longer necks survived and become new species. ➢ Environment selects the favourable variations in organisms

The diversity of beaks of finches in the Galapagos Islands



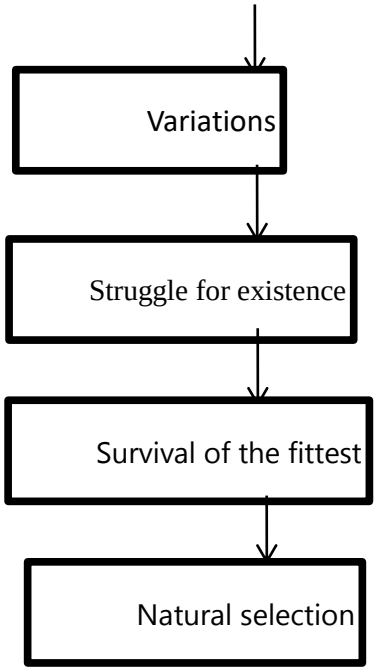
Evaluation:

1. What is the reason for the diversity of beaks in the finches of the Galapagos Islands?

Birds with beaks of suitable shape or size,depending on the availability of food resources in the environment, will survive and produce more offspring.

2.Haven't you understood that organisms with certain characteristics favourable to the environment tend to survive, reproduce and produce more offspring? Draw your conclusions by analysing Darwin’s explanation .





3.Complete the table

.....A.....	**DARWINISM**
The environment causes variations in organisms.	.....B.....

A. Lamarckism

B. The environment selects the favourable variations in organisms.

4. Which of the following is related to Lamarckism?

i)Natural selection ii)Struggle for existence

iii)Acquired variations iv)Survival of the fittest

a)i,ii b)i,iii c)iii Only d)iv Only

Answer : c

Neo Darwinism

- Discoveries of Gregor Mendel and the concepts about chromosomes and genes, it was recognised that the causes of variations that lead to evolution were **genetic changes, genetic recombination during sexual reproduction, and gene flow**.
- Darwinism became uncritically rationalised as more evidences and further studies from the fields of **population genetics, palaeontology, environmental science**, etc. were added to Darwinism to form Neo Darwinism.

Evaluation:

1.Neo Darwinism was formed by combining Darwin's evolutionary ideas with Mendel's discoveries.

a) What are the reasons for the variations that led to evolution?

b) What other scientific discoveries that contributed to the more evidence and further studies that led to Neo Darwinism ?

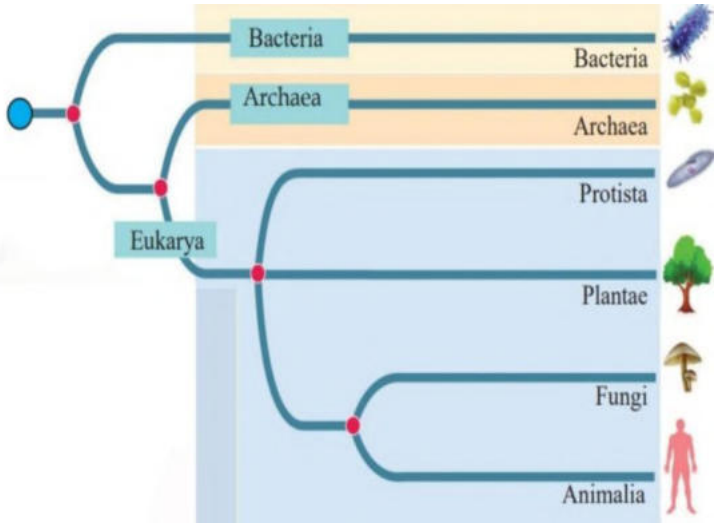
Ans: a) Genetic changes, genetic recombination, gene flow

b) Population genetics, paleontology, ecology ,environmental science.

Speciation

- Speciation is the process by which new species arise from a common ancestral species.
- LUCA - Last Universal Common Ancestors.
- MRCA - Most Recent Common Ancestor.

Evolution of different categories of organisms



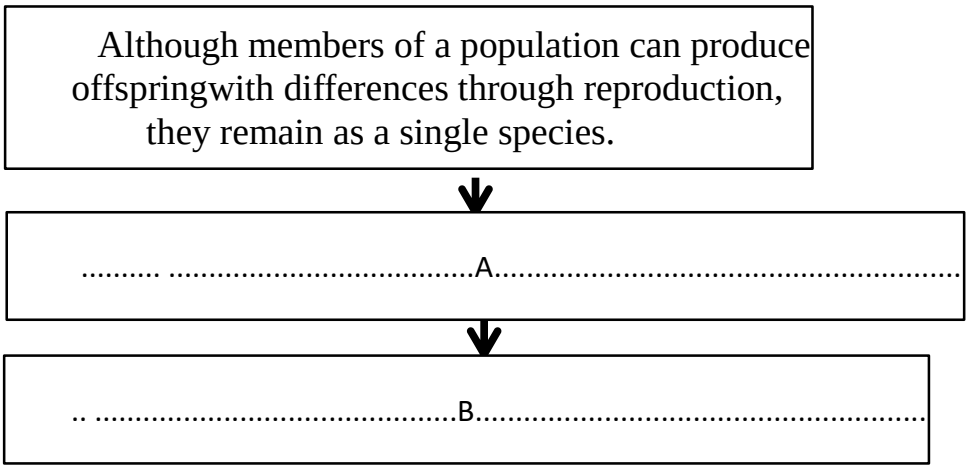
Modern research areas related to evolution

- Evolutionary Clinical Medicine
- Study of DNA
- Artificial Intelligence

Evaluation:

1. Biodiversity on Earth was originated through a process called speciation. Do you agree with this statement? Explain with reasons.

2.Complete the flowchart



A. If the members of a population get isolated from each other by ecological or other factors (mutation,natural selection, genetic recombination etc.) several variations might accumulate over time.

B. When the members of a species become unable to reproduce new offspring mutually, they will evolve into different species.

3.Choose the correct answer.

Statement 1: Speciation is the process by which new species arise from a common ancestral species.
Statement 2: LUCA – Most Common Ancestor.
Statement 3: Speciation is one of the reasons for biodiversity.

a) 1, 2, and 3 are correct. b) 1 and 2 are correct.

c) 2 and 3 are correct. d) 1 and 3 are correct.

Ans: d

Evidences of evolution

1.Molecular biology

The evolutionary relationship of organisms can be found out by comparing the sequence of nucleotides in the DNA and the sequence of amino acids in proteins in an organism with those of other organisms.

2.Comparitive Anatomy

The bones in the forelimbs of humans and cats, the flippers of whales and the wings of bats are similar. But, these organs differ in their external structure and function.

3.Fossils

- Fossils are the remains or traces of ancient organisms.
- Paleontology is the study of fossils.

Evaluation:

1.What are the conclusions that scientists have reached about how fossils support the process of evolution?

- Organic evolution is a gradual process.

  Eg: The ancestors of horses had shorter legs than the ones in out times.

- Fossils which are connecting links reveal the evolutionary relationship between organisms.

  Eg: Archaeopteryx that possesses the characteristic features of both reptiles and birds

- Many species have become extinct that were living on Earth in the past.

  e.g.: Dinosaurs, Mammoths

Primates

The common ancestors of mammals such as monkeys, apes, and humans belonged to the group of primates.

Features:

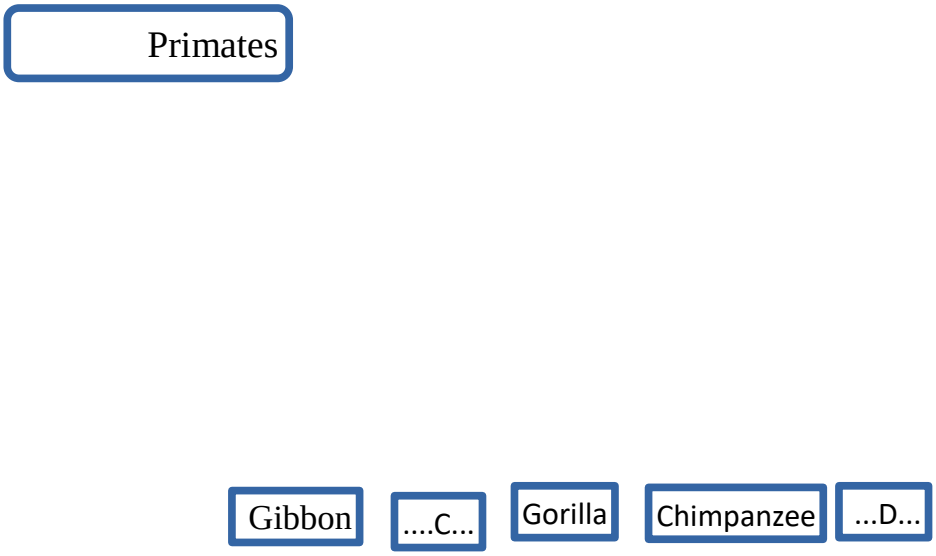
- A thump that can be opposed to other fingers,
- Binocular vision
- Relatively large and complicated brain
- Flat nails.

2. Did humans evolve from monkeys? Find out.

- Humans did not evolve from monkeys. Monkeys and humans both evolved as different branches from a common ancestor that belonged to the primate subgroup called Anthropoidea.
- From these, two main branches evolved.
- One branch is the Cercopithecoidea , which includes monkeys, and the other is the Hominoidea which includes apes and humans.

Evaluation:

2.Complete the Human evolutionary tree.



A.Small sized brain, Having tail B.Hominoidea
C.Orangutan D.Human

Members in the human evolutionary path

Human	Cranial capacity	Characteristics
Sahelanthropus tchadensis	350cm^3	• The first link in the human evolutionary series. • Fossils from Tchad in Africa
Astralopethecus	450cm^3	• Fossils from Africa • Bipedalism.
Homo habilis	600cm^3	• Fossils from Africa • Large skulls • Stones were used, Began huntining.
Homo erectus	900 cm^3	• Fossils from Africa, Asia and europe. • Walk upright on two legs. • Large foreheads. • Omnivores.
Homo neanderthalensis	1450 cm^3	• Fossils found from German.

		• Contemporaries of modern man. • Small, sloping foreheads. • Thick eye brows. • Buried dead bodies.
Homo sapiens	1350 cm^3	• Modern Man. • Acquired technology and agricultural methods. • Culturally most evolved, Built cities. • Domesticated animals.

Evaluation:

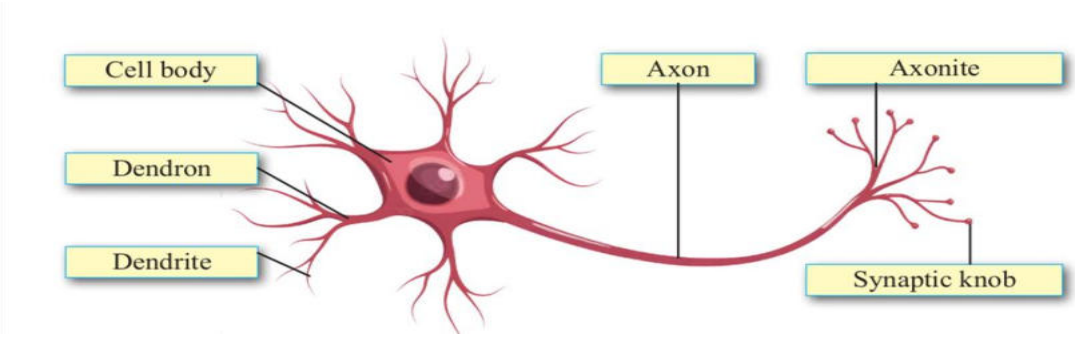
1. Complete the flowchart of the human evolutionary path..

Sahelanthropus tchadensis → ....A.... → Homo habilis → Homo erectus → .....B..... → .....C.....

The human nervous system

Parts: Brain, nerves, sensory organs, spinal cord

Structure of a neuron



Parts of a neuron and their functions.

Parts	Charecteristics	Functions
Dendrites.	Branch of dendron	Receive messages from adjacent neurons.
Dendrons	fine fibres arising from the cell body	Carries impulses from dendrite to the cell body.

Cell body	Cell body or cyton is the centre of the neuron. Cell membrane, cytoplasm, nucleus and cell organelles are seen here.	Passes impulses to the axon. .
Axon	The longest fibre from the cell body.	Carries impulses from the cell body to outside.
Axonites	Branch of axon.	Carries impulses to the synaptic knobs.
Synaptic knob	knob like structure seen at the tip of the axonite	Secretes neurotransmitter.

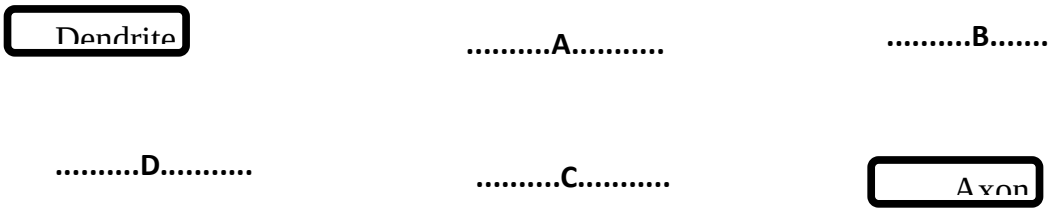
Evaluation:

1.Draw the structure of neuron and label the parts given below.

a.Branches of dendron

b.Part which secretes neurotransmitter.

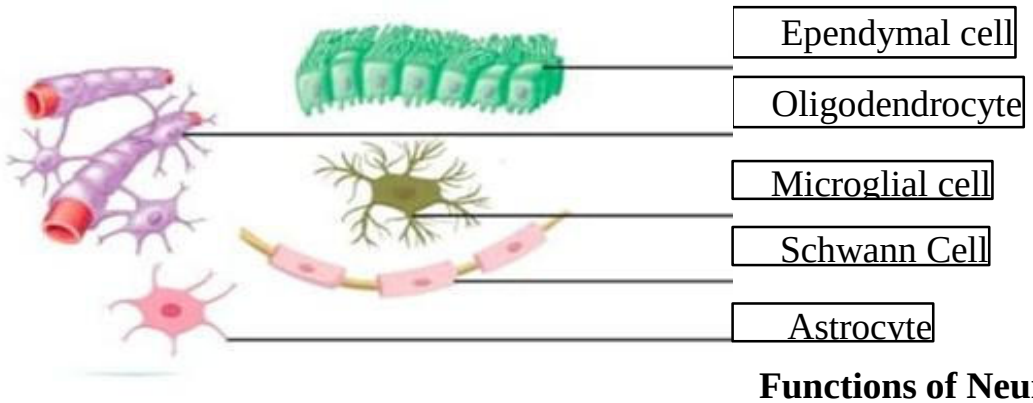
2.Complete the flow chart that determines the path of impulses through a neuron.



A.Dendron B.Cell body C.Axonite D. Synaptic knob

Neuroglial cells

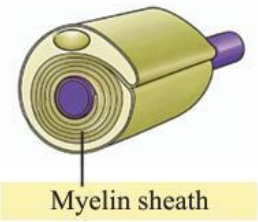
- More than half of the brain and spinal cord cells which have the ability to divide cannot receive stimuli or transmit messages.



Functions of Neuroglial cells

- Provides nourishment and eliminate wastes
- Protect from external injuries
- Increase the speed of impulses
- Act as defence cells
- Adjust pressure

Myelin sheath



In some neurons, the axon is covered by a layer called myelin sheath. It is made up of a shiny whi
Myelin.

The myelin sheath in the brain and spinal cord is made of oligodendrocytes, while the myelin sheath
made of Schwann cells.

Functions of Myelin sheath

- Provides nourishment.
- Protect the axon from external injuries.
- increase the speed of transmission of messages by acting as an insulator.

White Matter	Grey Matter
In the brain and the spinal cord, the part where myelinated neurons are more abundant is called the white matter	The part where the cell bodies and parts of the neurons without a myelin sheath are seen is called the grey matter.

Evaluation

1. Write one functional difference between neurons and neuroglial cells.

Cell	Function
Neuron	**.....A.....**
Neuroglial cells	**.....B.....**

A. Neurons have the ability to receive impulses.
B. Bring nutrition necessary for neurons, Eliminate wastes

2. Read the statements given below and answer the following questions.

1. The myelin sheath in the brain and spinal cord is made of oligodendrocytes.

2.The myelin sheath in the nerves is made of Schwann cells.

A) Statement 1 incorrect B) Statement 1 incorrect

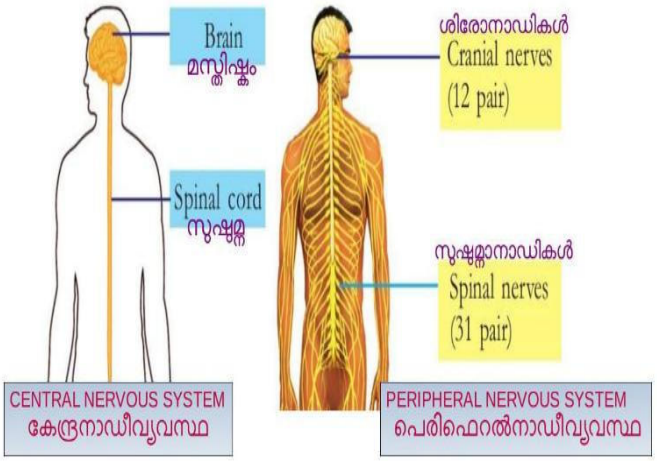
C) Statement 1and 2 incorrect D) Statement 1 and 2 correct.

Ans:D

Ganglia

The cyton parts of a group of neurons are covered by a membrane. These parts, seen spherical in shape.

Human Nervous System



Evaluation

1.

Nervous system

Brain Receptors

.....B..... Nerve ganglia

.....C.....

.....D.....

A. peripheral nervous system B. Spinal cord

C. Cranial nerves D. Spinal nerves

Protection of the brain and spinal cord

- Brain and spinal cord are covered by the three-layered Meninges.
- The cerebrospinal fluid is the fluid that is filled in between the inner membranes of the meninges, in the

cavities of the brain and in the central canal of the spinal cord.

Evaluation

1.The fluid found in the cavities of the brain and the central canal of the spinal cord is cerebrospinal fluid.

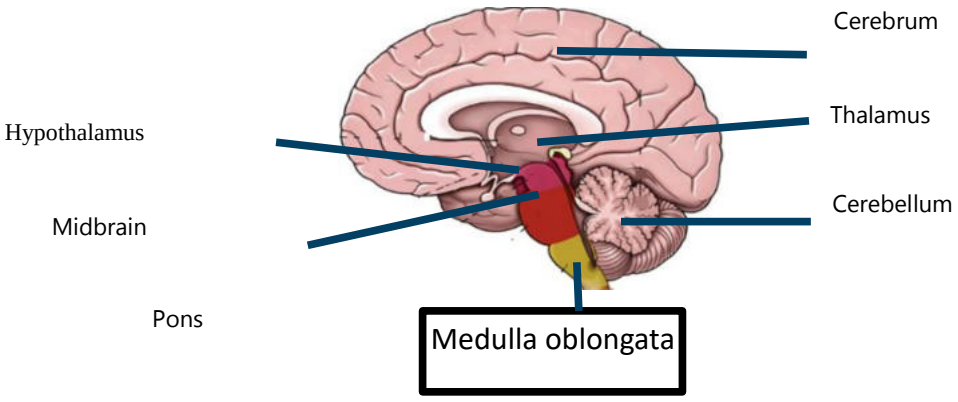
a) What are the functions of this fluid?

- Provides nutrition and oxygen to the tissues.
- Eliminates wastes.
- Protects from external injuries.

2.Which neuroglial cell helps in the production of cerebrospinal fluid?

Ans:Ependymal cells

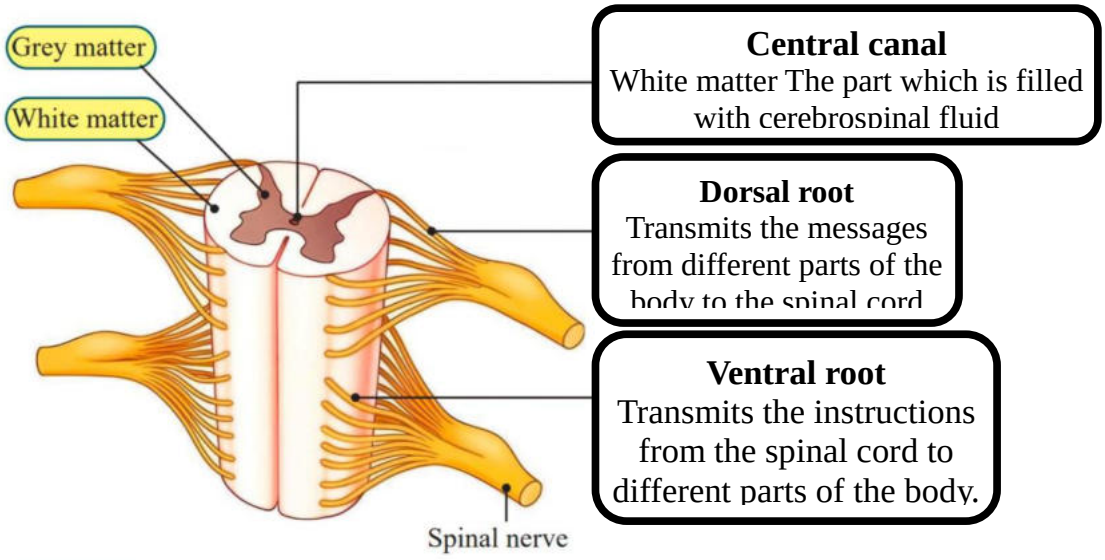
Human brain



Parts	Functions
Cerebrum	• Centre of problem solving, planning, memory, intelligence, thinking and imagination. • Voluntary movements. • Sensory experiences.

Cerebellum	• Maintain equilibrium of the body by coordinating muscular activities.
Thalamus	• Acts as the relay station of message. - Pain killers act here
Hypothalamus	• Maintaining homeostasis by regulating body temperature, hunger, thirst and emotions.
Medulla oblongata	• Controls involuntary activities like heartbeat, ventilation, vomiting, cough and sneezing.
Mid brain	• Primarily assess messages regarding vision and hearing. • A role in the movement of eyes and eye brows.
Pons	• Regulates the rate of ventilation. • Coordinates the muscular activities of the eye and the face.

Spinal cord

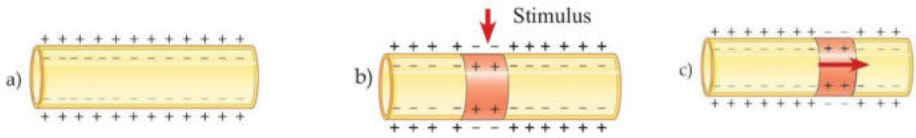


Evaluation

1. All spinal nerves mixed nerves.Why?

The spinal nerve is formed by the joining of the dorsal root, which carries messages from different parts of the body to the spinal cord, and the ventral root, which transmits instructions from the spinal cord to various parts of the body.

2. Understand the illustration of message transmission through a neuron given below and complete the column.



In the non-stimulated state, the outside of the cell membrane remains positively charged, and the inside remains negatively charged.

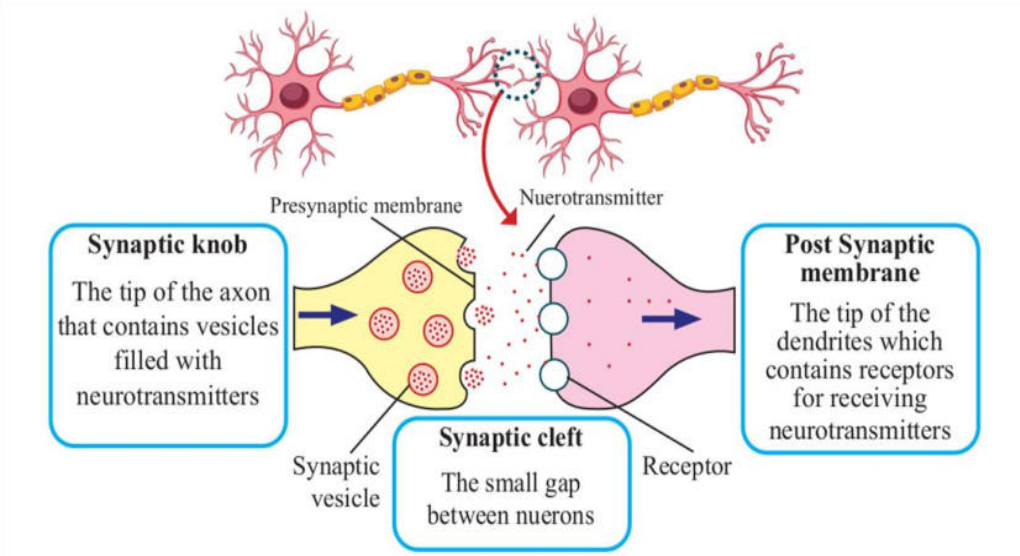
A. When stimulated, positively charged ions enter the cell through the cell membrane. This temporarily causes a change in charge in that region.

B. Opposite charges move forward. This causes the impulse to rapidly travel along the axon.

Synapse

- The part where an impulse is transferred from one neuron to another is called a synapse
- Synapses transmit the impulses to only one direction and increase the speed of the impulses.

The structure of synapse.



Evaluation

1. Fill up

Synaptic knob	**……A……**
……...B……..	The part of the dendrite that has receptors to receive neurotransmitters.

A. Post synaptic knob

B. The part containing vesicles filled with neurotransmitters.

Neocortex

- The cerebral cortex of the brain of mammals is modified into a complex structure of a six-layered neocortex.

- Advanced mental processes such as thinking, decision-making, learning, recalling, etc. are made possible with the help of the synapses of neocortex.

- When we learn new things or gain new experiences, the number of synapses increases, and the brain becomes more efficient.

Evaluation

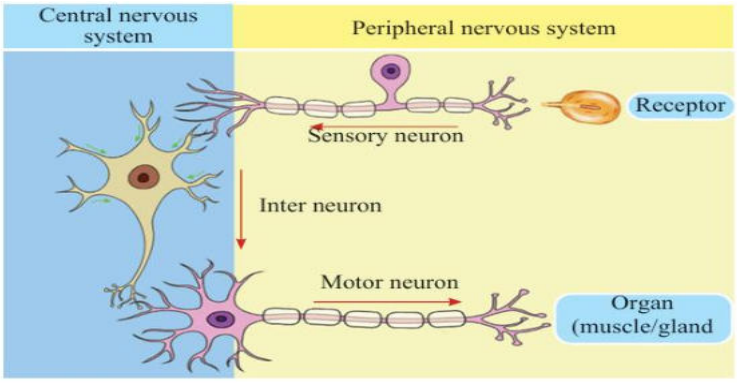
1.Although dolphins have a larger brain than that of humans, the level of thinking, language skills and social relationships are higher in humans. Evaluate this statement based on the following indicators.

A)Difference in the structure of the brain

b)Influence of natural selection.

Different types of Neurons

- Sensory neuron
- Motor neuron
- Inter neuron



Nerves

Nerves are made up of the axons of a group of neurons covered with a layer of fat and connective tissue.



Evaluation

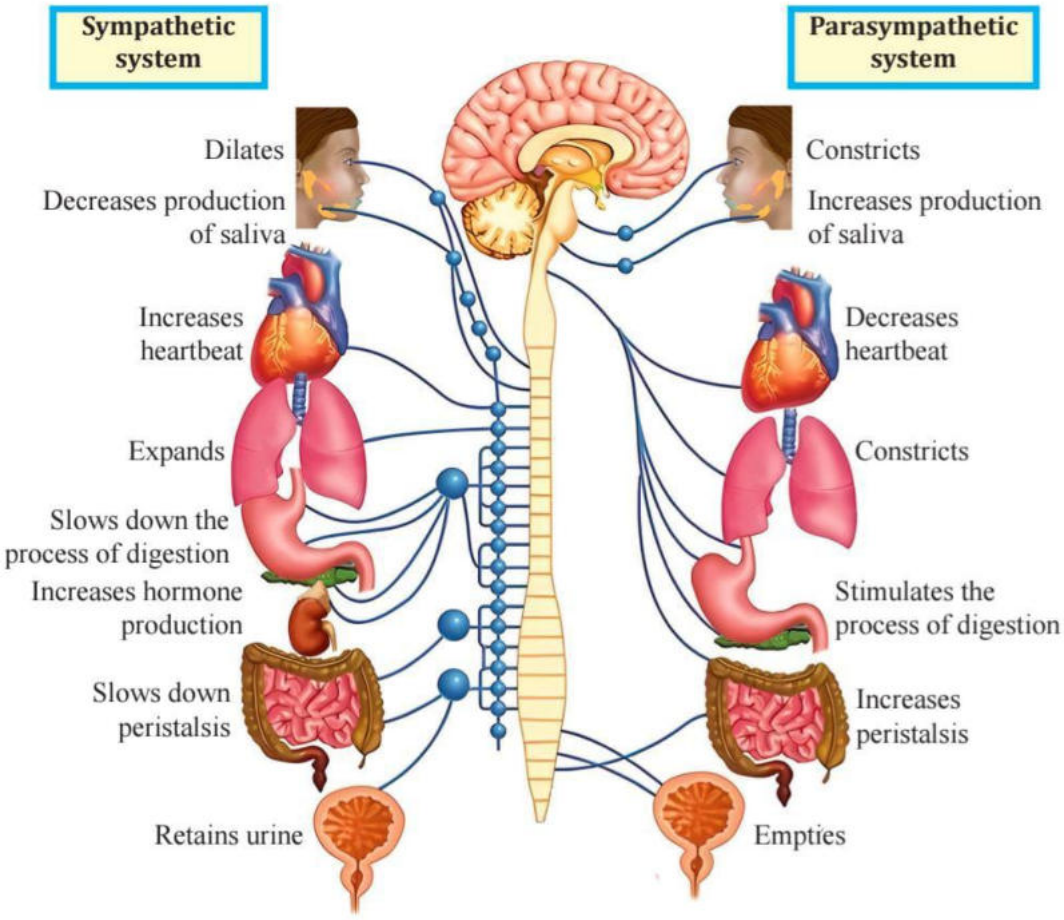
Nerve	Building block	Function
Sensory nerve	Sensory neuron	**.....A.....**
Motor nerve	**.....B.....**	Transmits instructions from the central nervous system to the organs.
Mixed nerve	Both sensory neuron and motor neuron.	**.....C.....**

A. It carries impulses from various organs to the central nervous system.

B. Motor neuronC. It carries sensory impulses to the central nervous system and motor impulses from the central nervous system to various organs.

Autonomous Nervous System

Autonomous nervous system is a part of the peripheral nervous system that regulates body activities by itself. It includes the sympathetic nervous system and the parasympathetic nervous system. Prepares the body to respond in an emergency situation.



Sympathetic system

Sympathetic system equips the body to respond during emergency situations.

Parasympathetic system

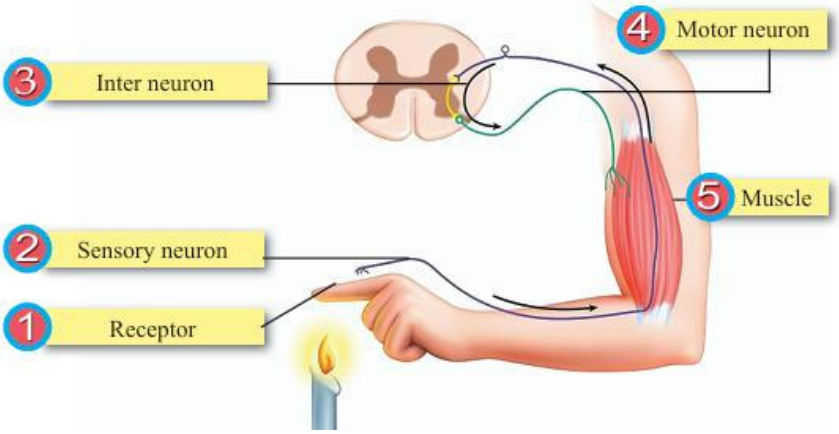
prepares the body to relax, and perform routine functions such as digestion.

Evaluation

1.The following are functions controlled by the autonomic nervous system. Provide an appropriate heading and list them.
Decreased saliva production, increased saliva production, dilated pupils, increased hormones, decreased hormones, urinary retention

Reflex action

- Reflex actions are reactions that occur spontaneously and involuntarily in response to stimuli.
- These actions can originate from both the spinal cord and the brain.
- The pathway through which impulses are transmitted in a reflex action is called a reflex arc.



Evaluation

Complete the flowchart

Receptor ⟶ .....A..... ⟶ Inter Neuron ⟶ .....**B.....** ⟶ .....C**.....**

A. Sensory Neuron
B. Motor Neuron
C. Muscle

The nervous system of different organisms

Hydra	Planaria	Insects
A neural network with no control center.	A pair of nerve ganglia.	• The nerve cells in the head region have merged and developed into the brain

		• The ganglia of paired nerve fibers emerging from this are seen in each segment.

Chapter 3
Behind Sensations

Stimuli and Responses

- Stimuli are the circumstances that lead to responses in living beings.
- Light, touch, heat, chemicals, etc. are examples of external stimuli.
- Thirst, hunger, pain, etc. are examples of internal stimuli.

Evaluation

1.Classify the statements given below and arrange them under appropriate headings.

Takes food, when hungry, A rabbit, upon seeing a lion, gets scared and runs away, Blanketing to keep out cold, Holding an umbrella to keep out the rain, Body temperature increases during certain diseases.

Situations	Stimuli	Responses
Takes food, when hungry	Hunger (Internal stimuli)	Eat food
A rabbit, upon seeing a lion, gets scared and runs away,	Seeing a lion (external stimuli)	Be scared.
Blanketing to keep out cold	Cold (external stimuli)	blanketing
Holding an umbrella to keep out the rain	Rain (external stimuli)	holding an umbrella
Body temperature increases during certain diseases	Disease (internal stimuli)	Body temperature is rising.

Receptors and Impulses

➔ Stimuli are recognized by the body through specialized cells or nerve endings. They are known as Sensory Receptors.

➔ Electrical impulses are produced in receptors in response to external and internal stimuli. These impulses are known as **receptor potential.**

➔ When such impulses are in higher concentration, **action potentia**l is formed

in the neurons associated with receptors.

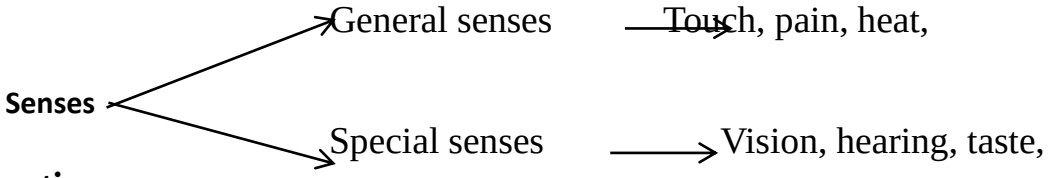
➔ Action potential travels through neuron as nerve impulses.

Evaluation



1. Identify A and B in the given diagram.
   A. Receptor potential in the receptors of the skin.
   B. Action potential in neuron.

Sensory receptors and senses



Evaluation

Complete the table below.

Sense organs	Receptors	Stimulus
Eye	Photoreceptor	.....A.....
.....B.....	Sound receptor	Sound
Tongue	.....C.....	Taste
.....D.....	Touch receptor Touch, cold, heat, pressure	.....D.....
Nose	.....E.....	Smell

A. Light B. Ear C. Chemoreceptors D. Touch, cold, heat, pressure
E. Olfactory receptors

Eye

- Position of the eye - Inside the eye socket
- The eye muscles hold the eyeball firmly within the eye socket and move it in different directions.

Conjunctiva

- Conjunctiva is the membrane that covers the anterior part of the eye, including the eyelids except the cornea.

Functions

- To protect the eye .
- To prevent dust, germs, and other substances from entering the eye.

- keep the eyes moist and lubricated.

Lacrimal glands

- These glands are present in the eyelids towards the upper part.
- Tears are produced by the lacrimal glands.

Evaluation

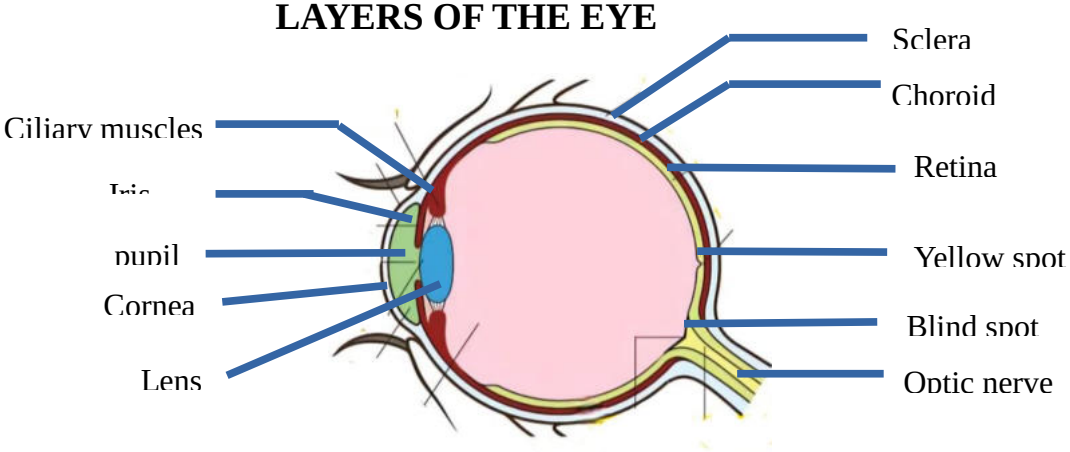
1. What are the functions of tears?

- For keeping the surface of the eye moist.
- Helps in providing nutrients and eliminate waste.
- It contains enzymes called lysozymes to protect the eye from infections.

2. Which enzyme helps to protect the eye from infection?

Lysozyme.

LAYERS OF THE EYE



Parts	Function
Cornea	Allows light to enrter the eye.
Ciliary muscles	Adjusts the curvature of the lens.
Iris	Regulate the size of the pupil according to the intensity of light.
Covex Lens	Forms a small, real and inverted image of the object on the retina.
Layer of photoreceptor	Photoreceptor cells (rods and cones) are found.
Bipolar cell layer	Transmits impulses from the photoreceptors to ganglion cells.
Ganglion cell layer	Transmits impulses from bipolar cells to the optic nerve.

Evaluation

1. On Bipolar cells

Bipolar Cell layer

.....P.....

A. There are two types of bipolar cell layers. Which bipolar cell layer is marked as P among them?

Off bipolar cell

B. What is the function of the bipolar cell layer?

Transmits impulses from the photoreceptors to ganglion cells.

2. Complete the flowchart provided, including the parts of the layers of eye and their functions.



A. Sclera B. Choroid C. Provides support and protection to the eye
D. Provides oxygen and nutrients to the layers of the eye

3. The iris is a part of the choroid layer.

A. Name the pigment present in the iris

Melanin

B. What are the functions of this pigment?

- Gives color to the iris.
- Absorbs ultraviolet rays.

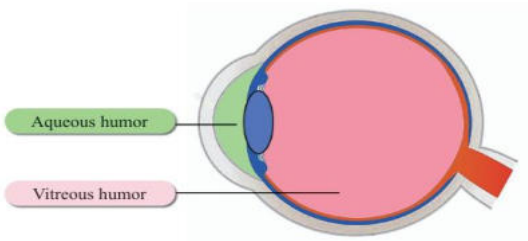
Blind spot

- There are no photoreceptor cells in the retina where the optic nerve originates.
- This part having no vision.

Yellow spot (Macula)

The yellow spot (macula) is seen in the middle of the retina where cone cells are abundant.

Eye Chambers and Humors.



	Features	Position	Function
Aqueous humor	Water-like fluid	Seen between the lens and cornea.	• Regulates the pressure in the aqueous humor. • Provides nutrients and oxygen to the lens and cornea.
Vitreous humor	Jelly – like fluid	Seen between the lens and retina.	• Maintains the shape of the eyeball.

Evaluation

1. Read the statements and choose the correct answer.

A. There are no photoreceptor cells in the retina where the optic nerve originates .

B. The part of the eye where there is no vision is called the blind spot.

C.Photoreceptor cells are found in the sclera.

a)A,B&C Correct b)A,B Correct, C Incorrect c)A,C Correct, B Incorrect C)A,B&C Incorrect

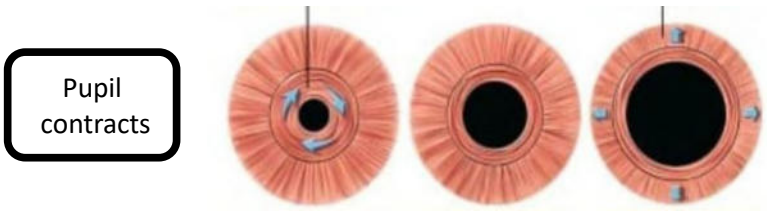
2. Complete the flowchart including the parts involved in the path through which light rays pass through the eye.

Cornea

Pupil

A.Aqueous humor B. Lens C. Retina

Muscular activity in the pupil



Evaluation

1. The size of the pupil is regulated by the radial muscles and circular muscles seen in the iris.

a) Write the differences between the radial and circular muscles of the eye in bright and dim light .

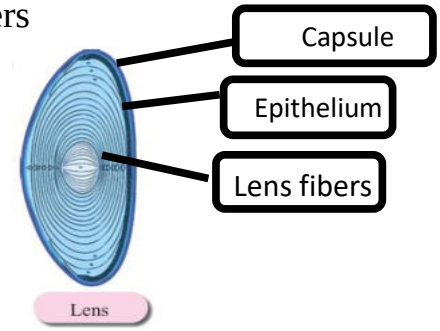
b)What is the difference in pupil size of the eye in bright and dim light?

Eye lens and ligaments

The main structural component of the lens is a protein called crystallin.

Parts of the lens

- Capsule (elastic membrane)
- Lens fibers



- Epithelium (produces lens fibers)

Ligaments

Lens got attached to the ciliary muscles through ligaments**.**

Power of accommodation

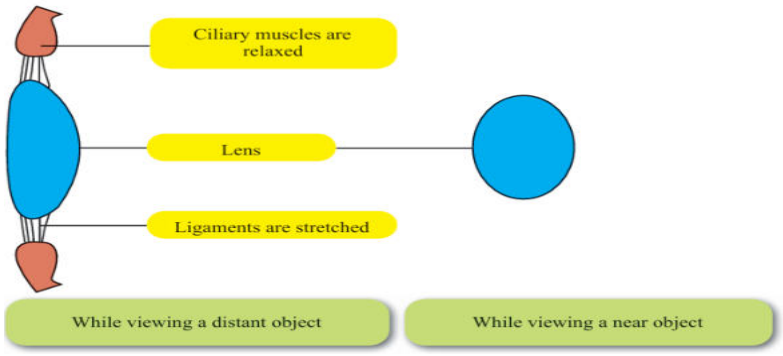
The ability of the eye to focus images of both near and distant objects accurately on the retina.

	Ciliary muscles	**Lens**	**Ligaments**
While viewing distant object.	Relaxed	Curvature decreases	Stretched
While viewing near object	Contracts	Curvature increases	Relaxes

Evaluation

1. Observe picture A and complete Image B.

A



B

Photoreceptor cells in the retina

Photoreceptor cell	Shape	Pigment	Function
Rod cell	Cylindrical shape	Rhodopsin	recognise objects in both dim light and in shades of black and white.
Cone cell	Cone shape	Photopsin	provide vision in intense light and also help in recognising colours.

- ➢ These two pigments are made up of a protein called opsin and a compound called retinal, which is formed from vitamin A.
- ➢ However, the chemical structure of retinal is different in rhodopsin and photopsin.

Evaluation



1. Identify P,Q in the figure.
2. Name the pigment and its components present in P,Q.
3. Which vitamin is necessary for the formation of pigments in photoreceptor?

Photoreceptors to the brain

<u>In the dark</u> **<u>In the presence of light</u>**

Colour Vision

- ➔ Colour vision is made possible when the three types of cones get stimulated in varying proportions when exposed to coloured light depending upon the intensity and wavelength of light.
- ➔ When red and green cones are stimulated together, the perception of yellow colour is formed.
- ➔ The stimulation of all the three types of cones creates the sensation of white light.
- ➔ The gene responsible for the production of pigments in cone cells which are sensitive to green and red are found in the X chromosome.
- ➔ The gene which is responsible for the production of blue cone pigment is found in chromosome 7.

Evaluation

1. Why are men more affected by colour blindness?

Because males have only one X chromosome.The gene responsible for the production of pigments in cone cells which are sensitive to green and red are found in the X chromosome.

2. Is retina necessary to see black colour? Find out.

It is impossible to see the color black in the absence of the retina because the experience of vision is possible only through the functioning of the retina, which contains photoreceptor cells.

Binocular fusion

The brain can integrate visual information from both eyes and recognize it as a single image. This is called binocular fusion.

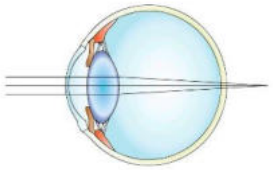
Causes and treatment methods of some eye diseases and disorders.

Eye diseases/ disorders	**Reason**	**Treatment**
Short-sightedness	Enlarged eyeball	• Spectacles with, Concave lens • Contact lens • Surgery
Long-sightedness	Eyeball becomes shorter Curvature of the lens decreases	• Convex lens • Contact lens • Surgery
Astigmatism	Irregular shape of the cornea	• Cylindrical lens

Cataract	Eye lens becomes opaque	• Surgery
Glaucoma	Failure in the reabsorption of aqueous humor pressure increases and the optic nerve gets damaged	• Laser treatment • Surgery
Conjunctivitis	Infection in the conjunctiva	• Medicines
Diabetic retinopathy	Uncontrolled diabetes	• Laser treatment • Surgery
Night blindness	Deficiency of vitamin A	• Eat food rich in vitamin A.
Xerophthalmia	Due to prolonged deficiency of Vitamin A, cornea becomes opaque	• Provides vitamin A • Corneal transplantation

Evaluation

1.



a) Name the eye defect indicated in the picture.
b) How can this be treated?

Protection of the eye

- Wash the eyes frequently, with clean water.
- Eat food rich in Vitamin A.
- While looking at screens like computers or mobile phones for a long time, blink your eyes at regular intervals.
- Ensure that you get good sleep, Avoid looking at intense/bright light.

Different methods used for testing eyesight

- ➢ Snellen Chart
- ➢ Retinoscope
- ➢ Tonometer
- ➢ Ishihara Plates

Evaluation

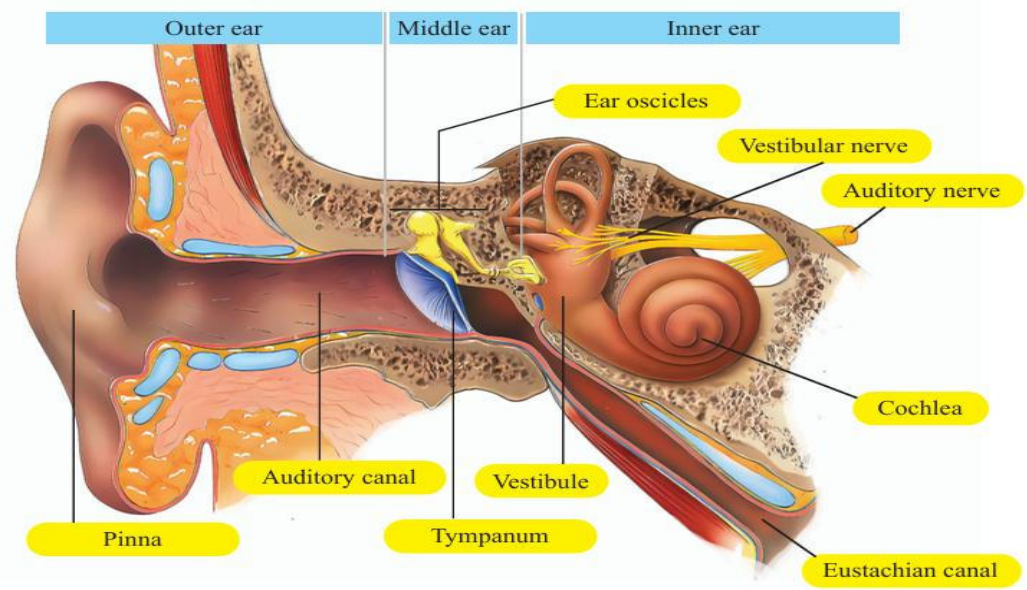
" Eye donation is a great donation". If eyes are donated after death, two blind

people can receive the gift of sight.

a)Which part of the eye is transplanted through eye donation?
Cornea

b) Prepare a poster based on the theme 'Eye donation is a great donation'.

Ear



The ear plays an important role in both hearing and maintaining body balance.

Functions of pinna

- Directs the sound waves into the auditory canal.
- Helps to identify the direction from which sound is produced.
- Protects the auditory canal to some extent from foreign particles.

Peculiarities of the auditory canal

- The auditory canal directs sound waves to the tympanum.

- Protects the tympanum from foreign particles.
- Prevents insects and dust particles from entering the ear.
- Ear wax found inside the auditory canal has disinfectant properties.

Tympanum/Ear Drum

- The tympanum vibrates in accordance with the sound waves.

Parts of the middle ear

1.**Ear ossicles (Malleus, Incus, Stapes)**

- Sound waves make the bones in the ear ossicles to vibrate.

2.**Eustachian canal**

- The eustachian canal is a long tube that connects the middle ear to the pharynx.
- It helps to balance the pressure on both sides of the tympanum.
- It facilitates the flow of mucus and fluids from middle ear to the pharynx.

The inner ear

Main parts: Cochlea, Vestibular system

The sense of hearing

- Sound waves vibrate the tympanum.
- This vibration passes through the ear ossicles and then vibrates the oval window.

Structure of Oval window

- The structure of the oval window is also similar to that of the tympanum.
- The cochlea has the shape of a snail's shell.
- It has 3 chambers.
- Oval window is the membrane that covers the opening towards the upper chamber

Evaluation

a. Auditory canal b. tympanum c. Incus
d. Stapes e. Cochlea f. Cerebrum

Evaluation

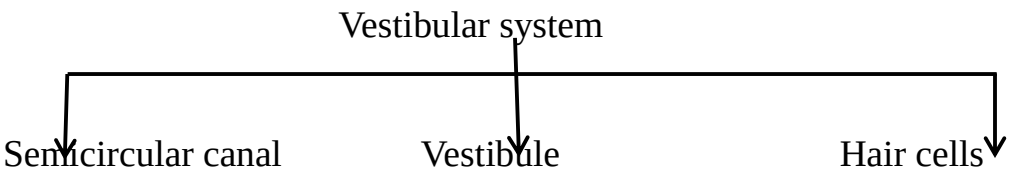
Parts related to hearing and their functions.

Part	Function
…..…...P…….	Carries sound waves to the auditory canal
Organ of Corti	..... Q .....
…….. R ……..	Carries impulses to brain
Eustachian canal	..... S.....

P. Pinna Q. Generates impulses
R. Auditory nerve S. Helps to balance the pressure on both sides of the tympanum

The inner ear

- The vestibular system is the part of the inner ear that helps maintain body balance**.**



- The endolymph present in the canals, which are arranged perpendicular to each other, moves with the rotational movement of the head.
- As a result of this movement,the hair cells present here get stimulated and impulses are formed.
- The utricle and saccule of the chamber called the vestibule also contain hair cells.
- The linear movement of the head generates impulses in these hair cells.
- When the impulses reach the brain through the vestibular nerve, the brain maintains body balance by receiving impulses from the eyes and muscles .

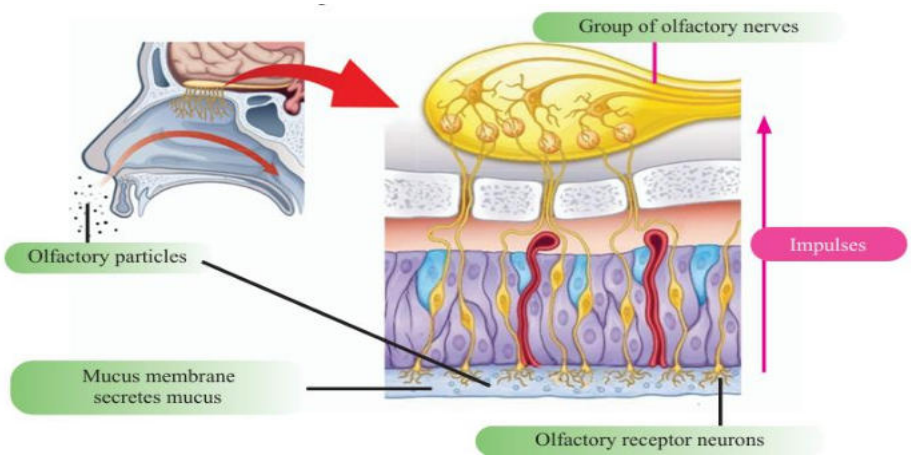
Choose and write the main functions performed by the ear from the options given below.

a. Helps to maintain body balance. b.Enables muscle movement.

c. Helps in hearing sound. d. Helps in smelling.

i) Both a and b are correct. ii) Both a and c are correct.
iii) Both b and d are correct. iv) Both a and d are correct.
Ans. ii

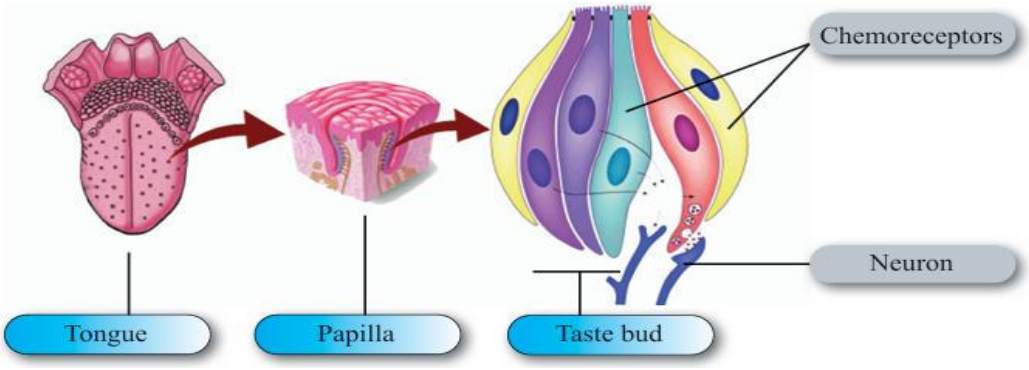
Olfaction



- While breathing, the particles responsible for smell enter the nasal cavity.
- It gets dissolved in the mucus produced by mucus membrane
- Millions of olfactory neurons in the mucus membrane get stimulated by special olfactory particles.
- Receptors generate impulses.
- They travel through the olfactory nerve to reach the part of the brain recognising smell.
- The sense of smell is effected.

Transmits impulses to the brain

Taste



A taste bud will have about 100 chemoreceptors. Microvilli from each chemoreceptor, reach the minute pores present in the papilla. Saliva enters through this pore.

How the sense of taste is created

a. The substances that give rise to taste are dissolved in saliva and reach the tiny pores in the papillae.

b.Stimulate chemoreceptors.

c.The chemoreceptor reach the brain through the nerve.

d. Make the sense of taste.

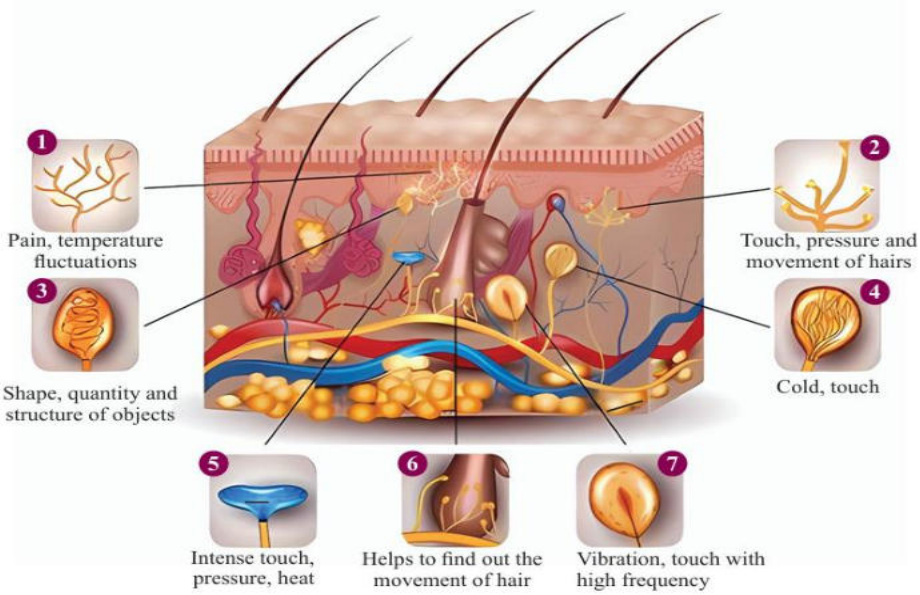
The main tastes we recognise are sweet, sour, salty, pungent, bitter and umami.

Evaluation

Taste	**Smell**	**Taste**
The fluid in which substances dissolve	**....1.....**	**.....2.....**
Stimulated receptor	**.....3.....**	**.....4.....**
The nerve that carries impulses to the brain	**.....5.....**	**.....6.....**

1. Mucus 2.Saliva 3.Olfactory receptor
4. Chemoreceptor 5.Olfactory Nerve 6. The nerve associated with taste

Skin - a sensory organ



Sensations that can be perceived through the skin

- Pain
- Temperature changes (heat, cold).
- Shape, size, and structure of objects.
- Touch, pressure.
- Movement of cilia.
- Vibration, high-frequency touch.

Evaluation

1. Which are the senses that can be perceived through the given part ?



Ans: Intense touch, pressure, heat

Sensory diversity in organisms

Organisms	Sensory system
Amoeba/Bacteria	Detects the presence of chemicals in the surroundings and move against them.
Euglena	The eyespot (stigma) helps to detect light and move towards it.

Insects	The compound eye made up of Ommatidia. The Antenna helps to detect smell and touch.
Bat	Ears and special type of echo location organ helps in hunting and travelling.
Hawk	Eyes with high vision, systems for long distance vision and detecting ultra violet radiation.
Snake	Jacobson's organ helps to detect smell.
Dog	Highly sensitive olfactory receptors are found.